

Harsh Lara

☎ 412-583-1116 | ✉ hlara.tech@gmail.com | 🏠 harshlara.github.io | 🌐 harshlara

Work Experience

Resolve AI

San Francisco, CA

Sept 2025 – Present

MEMBER OF TECHNICAL STAFF, RESOLVE AI

- Built a horizontal memory architecture supporting multi-turn, agentic workflows across the [resolve.ai](#)
- Developing a code-generation remediation agent that significantly reduces incident mean-time-to-resolve (MTTR)

Google DeepMind

Mountain View, CA

Sept 2021 - Present

RESEARCH ENGINEER, GEMINI

- Led post-training for improving tool-use capabilities in [Project Astra](#) and [Gemini Live](#), enabling critical integration with Google Workspace and significantly enhancing productivity for users
- Co-led the design and implementation of [MoDE](#): A multi-task parameter efficient fine-tuning technique, outperforming existing methods and achieving significant performance improvements across over 700 tasks ([published at NAACL 2025](#))
- Core contributor to [OmegaPRM](#): A novel MCTS algorithm for automated process supervision, directly improving mathematical reasoning in Gemini models from 51% to 69.4% on MATH500 and 86.4% to 93.6% on GSM8K
- Core contributor to [RecLLM](#): LLM-based Conversational Recommender System (CRS),

Microsoft

Bellevue, WA

Dec 2019 - Sept 2021

APPLIED SCIENTIST, AZURE MACHINE LEARNING

- Led the design and implementation of end-to-end neural network infrastructure, significantly improving efficiency for text data-labeling training and inference processes.
- Collaborated directly with key enterprise customers to architect and deliver tailored AutoML solutions, driving an increase of over \$0.5M in Annual Recurring Revenue (ARR)
- Engineered a model wrapper achieving 95% reduction in model size, optimizing resource utilization and deployment efficiency

Apple

Cupertino, CA

Summer 2019

MACHINE LEARNING ENGINEER INTERN, SIRI

- Developed an innovative method to dynamically interpolate probabilistic language-models on-device achieving multi-domain increase (>15%) in ASR accuracy. Received strong recognition from senior AI/ML leads.

Education

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

MS, COMPUTATIONAL DATA SCIENCE

- Selected courses: Cloud Computing | Computer Systems | Intro to Machine Learning | Language and Statistics | Large-Scale Multimedia Analysis | Applied Machine Learning | Deep Learning | Computational Ethics for NLP

National Institute of Technology Karnataka (NITK)

Surathkal, India

B TECH, COMPUTER SCIENCE AND ENGINEERING

- Selected courses: Operating Systems | Databases | Distributed Systems | Object Oriented Programming | Advanced Algorithms | Data Structures | Image Processing | Data Communication | Graph Theory | Discrete Math | Linear Algebra

Skills

Strengths: Artificial Intelligence, Language Technologies, Machine Learning, Deep Learning, Cloud Computing

Languages/Libraries: Python (Advanced), C/C++ (Advanced), Java (Intermediate), SQL (Intermediate)

Tools/Technologies: PyTorch, Tensorflow, GCP, Azure, AWS

Projects

Effective Multi-task Parameter Efficient Fine-Tuning of LLMs

Google DeepMind

- Developed a parameter-efficient framework for multi-task adaptation in LLMs
- Designed a novel task-specific adapter architecture that outperforms existing methods in multi-task fine-tuning
- Demonstrated significant performance improvements in handling over 700 tasks while maintaining computational efficiency
- Contributed to advancing lightweight, high-performance models for a wide range of applications

Improve Mathematical Reasoning in LLMs by Automated Process Supervision

Google DeepMind

- Developed OmegaPRM, an efficient MCTS algorithm that automates the collection of high-quality process supervision data
- Engineered a divide-and-conquer approach within OmegaPRM to swiftly identify errors in Chain-of-Thought reasoning,
- Collected over 1.5 million process supervision annotations, training Process Reward Models, significantly boosting LLM performance on complex mathematical reasoning
- Improved LLM performance from 51% to 69.4% on MATH500 and from 86.4% to 93.6% on GSM8K

Leveraging LLMs in Conversational Recommender Systems

Google Research

- Developed RecLLM, a conversational recommender system for YouTube videos utilizing LaMDA
- Designed an architecture integrating LLMs to improve dialogue management, and provide explainable recommendations
- Developed a controllable LLM-based user simulator to generate synthetic data, addressing data scarcity challenges
- Demonstrated RecLLM's fluency and diverse functionalities through illustrative conversations, showcasing its potential in personalized recommendation systems

Publications

- [MoDE: Effective Multi-task Parameter Efficient Fine-Tuning with a Mixture of Dyadic Experts](#), NAACL 2025
- [Improve Mathematical Reasoning in Language Models by Automated Process Supervision](#), arXiv 2024
- [Leveraging Large Language Models in Conversational Recommender Systems](#), arXiv 2023
- [Evaluation of Synthetic Datasets for Conversational Recommender Systems](#), NeurIPS 2022